Tutorial 5-6 (final project) Model calibration, ESILV, 2025-2026

Work to be submitted before 8pm on Saturday 24 January 2026. To be done in groups of two students of the same TD group (with the possibility of having a group of 1 if there is an odd number of students, but not a group of 3). Create your group and indicate its composition in this form (only the one among the two of you who you consider to be the "corresponding" student has to give an answer), before Friday 19 December at 1pm: https://forms.gle/wedpiwYEMeWVBpYZ9

For the submission of the project, you will send by email to Matthieu Garcin a single pdf file of **20 pages maximum (appendices included)** containing your answers to each question, with the graphs that you have generated and commented on, as well as the **code directly inserted in the same pdf** (a pdf generated in LaTeX will be appreciated positively). Analyze and comment on your results for each question.

Any plagiarism, use of LLM or any software automatically generating text, code or images is dishonest for an academic work and is therefore strictly prohibited. Any violation of this rule for any part of your work may result in a grade of 0 for the entire project (and possibly a disciplinary council). The use of libraries answering the questions of this project is also prohibited.

For the entire project, we are given European call prices of one-year maturity, all defined on the same underlying whose spot price is 100€:

Strike	95€	96€	97€	98€	99€	100€	101€	102€	103€	104€
Price	10.93€	9.55€	8.28€	7.40€	6.86€	6.58€	6.52€	6.49€	6.47€	6.46€

If needed, call prices of other maturities will be assumed to be such that their implied volatility is the same as the one of the call of same strike in the above table. If needed as well, European put prices can be obtained from call prices of same strike and maturity, using the no-arbitrage hypothesis. The risk-free rate is 0.2%.

I – Focus on volatility

1/ Using the above dataset, calibrate the implied Black-Scholes volatility for all the options.

2/ Using the same method as the VIX (except regarding maturities), calculate a volatility index consistent with the dataset displayed above and compare it to the volatilities of question 1.

3/ Using the above dataset of one-year maturity, estimate the Heston model explaining in average the price of all the options (you must be able first to price an option with this model).

4/ Calibrate (and not estimate) Heston model only on the three options the closest to the money.

5/ Estimate a single Heston model explaining in average the price of all the options of maturity 6 months, one year, 18 months (one model for the three maturities together).

6/ How are the parameters affected by the different settings in questions 3, 4, and 5? For each of the three sets of parameters obtained for Heston model, price a one-year Asian option of strike 98€, the average appearing in the payoff being calculated as a discrete average of daily prices between the issue

date (the option is issued today) and maturity. Comment (try to justify) the difference between these prices and the price of the European call of same strike and maturity.

II – Non-Gaussian alternatives

7/ Estimate the SVI model to depict in average the volatility skew obtained in question 1.

8/ Using Breeden-Litzenberger formula, provide the risk-neutral probability of the underlying price in one year. Use the SVI model estimated above to enrich your dataset first. Quantify the divergence between the risk-neutral distribution and the lognormal distribution of the Black-Scholes model.

9/ Using the historical time series of the underlying provided apart, find a few relevant empirical properties that make the dynamic different from a geometric Brownian motion and write the corresponding model of dynamic. The relevance, the rigor, and the creativity of the answer will be assessed.

10/ Using your model defined in question 9, calibrate a weighted Monte Carlo on the dataset of 10 European call prices. Use this calibration to price the Asian option of question 6. Comment on the result.